ADVANCED PROBLEMS FOR LECTURE 12

In the lecture, we will be talking at a more general level of the patterns of what is going on. But there are deeper questions to explore therein, which we cannot get to. This is your opportunity to investigate deeper for yourself. Don't limit yourself to just these questions, let your curiosity guide you too! That is, if you wonder about any details, or if any certain patterns or ideas exist, see if you can state your question clearly, and answer it! Here are a few suggestions to get you started.

Let's make some useful definitions.

Definition 0.1. We say a function f is *increasing* on a set S if for any $x, y \in S$ with $x \leq y$, we have $f(x) \leq f(y)$.

We say f is *strictly increasing* on S if whenever $x < y$ we have $f(x) < f(y)$.

(1) Is it true that if f is increasing on S and increasing on T, then f is increasing on $S \cup T$?

(2) We like to jump on the derivative train and say that positive derivative implies increasing. This makes some intuitive sense, but how do we prove this rigorously? That is, if a function is differentiable on (a, b) with $f'(x) > 0$ for all $x \in (a, b)$, show that f is (strictly) increasing on (a, b). (This is not trivial. See a hint/approach on the back of the page if you find yourself getting too stuck.)

(3) What about the converse of this statement? Is it true that if a function is strictly increasing on a set (a, b), then $f'(x) > 0$ for all $x \in (a, b)$?

(4) Suppose we have a differentiable function f, and the only zeroes for its derivative are a and b where $a < b$. Justify rigorously (by appealing to a named result) why we only need to test one point $c \in (a, b)$ to determine the sign of the derivative over the whole interval (a, b).

Look up the "Cantor Function" (also called the devil's staircase). Here we have a continuous function which has many spots where it is not differentiable (it is not differentiable on the Cantor set). But the places where it is differentiable, it's derivative is 0. Yet somehow it is rising from 0 to 1. Strange!

Hint for question 2: This question is not easy. We likely need to build up something called the "Mean Value Theorem". This result states that

Theorem 0.2. *If f is continuous on $[a, b]$, and differentiable on (a, b), then there is some $c \in (a, b)$ such that*

$$f(b) - f(a) = f'(c)(b - a).$$

So, can you see how this result implies the statement in question 2? But now of course the question remains on how to prove this. I recommend the stepping stone of Rolle's theorem, which states:

Theorem 0.3. *If f is continuous on $[a, b]$ and differentiable on (a, b), and $f(a) = f(b)$, then there is some point $c \in (a, b)$ such that $f'(c) = 0$.*

And... how to prove this? Try using a result from Lecture 11 (last time).